Process-Oriented Hazard Analysis for AI Systems

Overview

This document details a set of guidelines for applying System Theoretic Process Analysis (STPA)[1,2] for algorithmic systems that are based on machine learning (ML) algorithms. We call this Process-oriented Hazard Analysis for AI Systems (PHASE). STPA is a well-established analytical framework from the field of safety engineering developed by Nancy Leveson. It takes a control and system theoretic approach to identify key losses and hazards to be avoided and examines how elements of a system and their interactions relate to the given losses and hazards (Leveson 2016). Traditionally, STPA was developed for complex safety-critical systems whose failure could result in loss of life, significant property damage, or damage to the environment (Knight 2002)[3]. PHASE illustrates a way of interpreting STPA concepts and steps for ML-based algorithmic systems, recognizing that many of the harms from deployment of these systems go beyond traditional safety-critical losses such as loss of life.

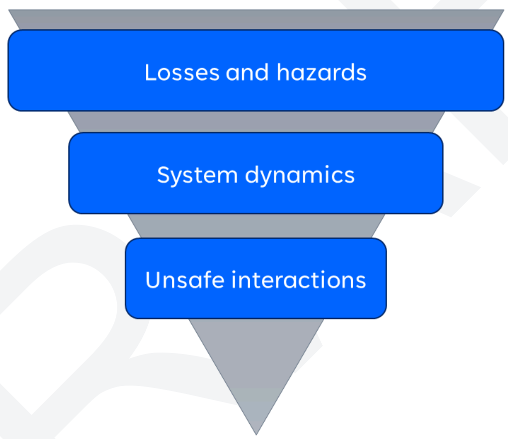


Figure 1. Flow of analysis for PHASE is modelled after the four key steps in STPA

What is in this guideline?

PHASE elaborates on the four stages of the STPA framework, which broadly include 1) Identifying losses and hazards, 2) Mapping out system dynamics, 3) Examining unsafe interactions, and 4) Identifying potential causal scenarios for these unsafe interactions. Next, we discuss the reasons for creating this guideline and how it was developed. We also discuss remarks about the mechanics (i.e.,

[1] Leveson, N., & Thompson, J. (2018). *STPA Handbook*. https://psas.scripts.mit.edu/home/get_file.php?name=STPA_handbook.pdf
[2] Leveson, N. (2011). *Engineering a Safer World: Systems Thinking Applied to Safety*. MIT Press.
[3] Knight, J. C. (2002). Safety critical systems: challenges and directions. *Proceedings of the 24th International Conference on Software Engineering. ICSE 2002*, 547–550.

who? when?) of applying PHASE. The remainder of this document describes the interpretation of these steps formally (with an accompanying example for the COMPAS algorithm).

Why did we create PHASE?

1) **Bringing a Control-Theoretic Perspective to ML Systems**
Machine learning (ML) systems are inherently probabilistic, yet they operate within structured interactions where different sub-systems influence each other. PHASE applies control theory to ML-based systems, framing them as networks of interconnected components—such as models, datasets, and human actors—that either control or are controlled by other sub-systems. Even though ML models are probabilistic, this perspective helps us analyze how different components affect system behavior.
This guideline provides a step-by-step approach to applying the Systems-Theoretic Process Analysis (STPA) framework to ML-based systems, emphasizing their complexity and the interactions between automated, algorithmic, and human sub-systems. Additionally, it highlights the evolving nature of ML systems, which change over time based on factors such as input data, model architecture, and usage patterns. PHASE equips practitioners with the necessary analytical framing to account for these dynamics.

2) **Addressing a Broader Range of Harms Beyond Safety-Critical Failures**
Many algorithmic systems operate in safety-critical contexts where failures can result in physical harm or damage to property and the environment. However, the risks of ML-based systems extend beyond traditional safety concerns. In recent years, media reports and academic research have documented various adverse impacts, including discrimination against marginalized groups, the spread of misinformation, and privacy violations.

While existing STPA applications primarily focus on safety-critical domains, the STPA handbook acknowledges the framework's potential for analyzing a wider range of harms. PHASE serves as a guideline for applying STPA to ML-based systems in ways that go beyond traditional safety risks, enabling a more comprehensive hazard analysis that accounts for social and ethical concerns.

3) **Making STPA Accessible to ML Practitioners**
The original terminology and concepts in STPA can be difficult to navigate for ML practitioners such as product managers, responsible AI analysts, and ML engineers/researchers [ref]. PHASE bridges this gap by translating STPA concepts into a more accessible framework, ensuring that practitioners can effectively understand and apply the methodology in their work.

By providing this structured guideline, PHASE makes it easier for ML professionals to conduct hazard analyses, assess risks, and design more accountable and reliable ML-based systems.

How did we develop PHASE?

To develop PHASE, we followed a structured, iterative process that combined industry insights, comparative analysis, and real-world case studies. Here are the key steps we took:

1) **Understanding Industry Practices and Practitioner Perspectives**
   We began by examining current ethical and social risk management practices used in the industry. We also gathered insights from ML practitioners about their experiences with safety engineering frameworks such as Failure Modes and Effects Analysis (FMEA) and Systems-Theoretic Process Analysis (STPA). Practitioners particularly valued STPA's focus on starting with harms and its ability to map out complex system dynamics.
2) **Comparative Analysis of FMEA and STPA for Text-to-Image Models**
   We conducted a comparative analysis of FMEA and STPA in the context of text-to-image models used in creative practice. This analysis highlighted that STPA offers a more structured approach to identifying potential sources of harm and accounts for hazards arising from contextual factors, such as how a company structures its development processes and how an ML system is embedded in a product. We followed the STPA methodology as outlined in the Leveson and Thomas handbook.
3) **Applying STPA to Additional Case Studies**
   Beyond text-to-image models, we applied STPA to two additional case studies:
   a) The use of predictive algorithms in clinical decision-making.
   b) The use of reinforcement learning (RL) algorithms for determining injection dosages in artificial pancreas systems.

   These applications helped refine our understanding of STPA's strengths and informed the first iteration of PHASE.

4) **Piloting PHASE Through Teaching and Workshops**
   We tested the first version of PHASE by teaching it to students in a graduate-level course on the Ethics of Intelligent Systems. We also introduced it in workshops and tutorials, including the Sociotechnical Safety Workshop at HAII Stanford and FAccT 2024. These pilots provided valuable feedback that helped us further refine the framework.

Through this process, PHASE evolved into a structured guideline tailored for ML practitioners, bridging the gap between safety engineering principles and the unique challenges of ML-based systems.

How should you use this guideline document?

We recommend that you use PHASE and the supplemental case studies along with established STPA resources such as the STPA Handbook and instructional videos provided by the creators of STPA.
[Maybe add something about setting up PHASE on GitHub as an open-source guideline]

This type of hazard analysis can and should ideally be conducted early in developing an ML-based product such that discoveries from the process could inform future design requirements and decisions. Considering the changing nature of ML-based products, it is important to update the analysis, especially if there are changes in the control structure diagram.

The individuals who conduct this analysis need to have (or be able to gain) a global understanding of how the ML system works and who is involved in developing/launching it. Having a team of interdisciplinary experts trained in the PHASE guideline and STPA framework collaborate on the analysis is highly beneficial. Their diverse perspectives enhance every step of the process, from identifying losses and defining system boundaries to mapping the system and uncovering potential causal scenarios.

Step 1: Identify the purpose of the analysis

Identify losses and hazards

PHASE framework draws attention to the social context, the adverse impact of an ML-based algorithmic system, and potential sources of harm to identify the purpose of analysis. To identify the purpose of analysis, follow the key definitions and steps to 1) Identify losses and 2) Identify system-level hazards.

Identify losses

STPA starts with identifying *losses*, which is defined broadly as the loss of something stakeholders consider valuable. For instance, the loss of a person's life would be important in the case of analyzing a car as a system. To identify losses, start by *identifying the key stakeholders and understanding their values*. However, there is an extensive range of possible stakeholder values, and all value losses are not necessarily problematic. To develop and refine the list of losses:

- Examine if any of the common safety-critical losses are applicable to the case. These include loss of human life or human injury, property damage, and the environment. If so, enlist them as potential losses.
- Examine if performance-related losses are applicable here. Performance-related losses emerge when the ML system does not meet commonly used and desired performance metrics such as accuracy, recall, and AUC (area under the curve) value. For ML systems, performance losses could include loss of user engagement, loss of credibility, loss of public trust, and loss of venture capital. [4]

[4] This list of performance-related losses is only illustrative, and it is not comprehensive across all ML applications.

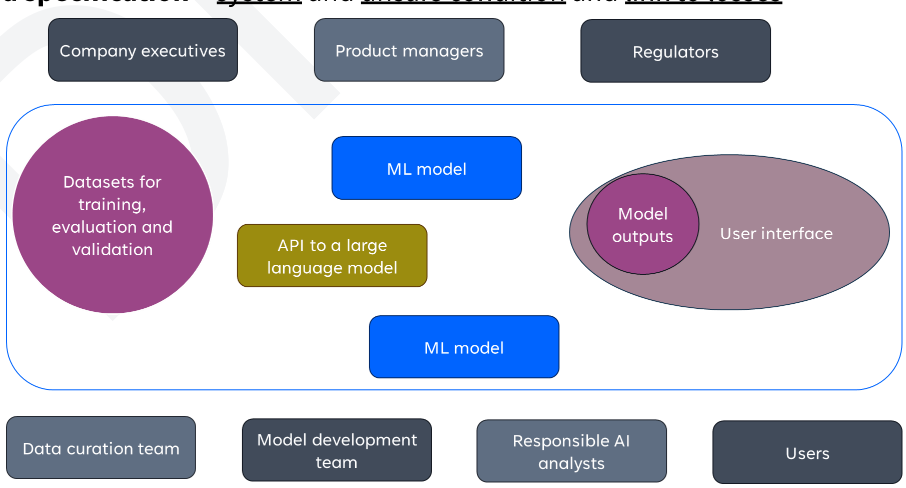
- Examine losses from ML-based systems that are not covered by safety-critical and performance-based losses. As mentioned earlier, harms from ML-based systems expand beyond inaccurate outputs and physical injury. Sociotechnical harms from algorithmic systems occur when the use of these systems leads to losses such as loss of justice, loss of the right to privacy, and loss of sense of identity. To identify such losses:
    - Use existing credible algorithmic harm taxonomies as a method of identifying potential losses. For example, the taxonomy of sociotechnical harms by Shelby et. al, outlines five categories of harms, including representational, allocative, quality-of-service, interpersonal, and social system harms (Shelby et al. 2023)[5].
    - Re-examine and integrate the stakeholder values (identified earlier) that are critical to include, as loss of such values could result in harm to impacted individuals and groups
    - Review and monitor existing regulatory requirements for losses that are important to minimize/avoid by regulatory standards.

The final list of losses generally contains 4-6 items, and it should not exceed 10 losses.

Identify system-level hazards

Once losses are identified, the next step is to identify hazards. Before outlining the necessary steps to identify hazards, here are some key terms to help you with successful hazard identification (Leveson 2016):

- **System definition:** A system is a set of components that act together as a whole to achieve some common goal, objective, or end. A system may contain subsystems and may also be part of a larger system.
- **Hazard definition:** A hazard is a system state or set of conditions that, together with a particular set of worst-case environmental conditions, will lead to a loss.
- **Hazard specification** = system and unsafe condition and link to losses



[5] Shelby, R., Rismani, S., Henne, K., Moon, A., Rostamzadeh, N., Nicholas, P., Yilla-Akbari, N. 'mah, Gallegos, J., Smart, A., Garcia, E., & Virk, G. (2023). Sociotechnical Harms of Algorithmic Systems: Scoping a Taxonomy for Harm Reduction. *Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society*, 723–741.

Figure 2. ML-based systems are complex and consist of many algorithmic, automated, and human sub-systems. This figure shows some of the sub-systems (technical and human) involved in ML development.

To identify system-level hazards:

- Identify system boundaries: This also defines the scope of the analysis. Select the system boundary based on three factors:
    - The boundary should include the parts of the system that its designers have some control over
    - The amount of information you have about parts of the system as the analyst
    - The stated/necessary goal of the analysis based on existing regulations/policies

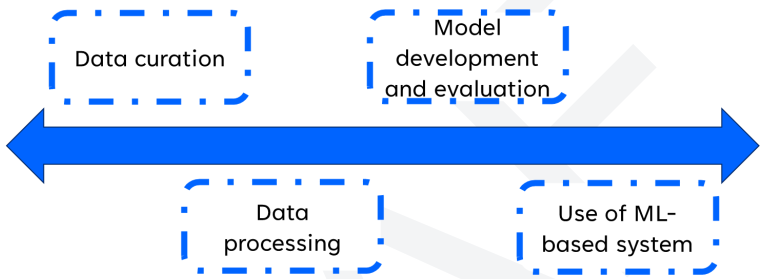


Figure 3. Examples of potential system boundaries along the ML development pipeline

For algorithmic systems, the part of the ML development process that system designers have control over includes data processing protocols for curating train/validation/test datasets, evaluation protocols for the model and its output, model architecture development, infrastructure for setting up necessary compute resources and training process, integration of different ML models in a given product, and user interaction design affordances. Once a system boundary is identified,

- Brainstorm about the hazards based on two pieces of information: the losses and the overall objective of the chosen system. Consider what system state and conditions would lead to loss X, given the objective of the system. For example, if we are looking at the scope of data processing, the objective of the system is to create a training dataset. For loss of privacy, a potential system-level hazard would be: *the training dataset is not anonymized*, or *the training dataset breaches privacy guidelines as outlined by a specific privacy guideline*. For loss of justice, a potential system-level hazard would be *the training dataset only contains data from a specific stakeholder group* OR *the training dataset is not diverse in representation*.

A list of 2-6 system-level hazards is a good place to start. Hazards need to be connected to the list of losses created in the first part of this analysis.

Example:

Losses:

- L1: loss of reputation (for Northpointe) - (performance related)
- L2: loss of justice (representational and allocative harms)
- L3: loss of credibility for judge (interpersonal harm)
- L4: loss of privacy (interpersonal harm)

System boundary:

- S1: Use of COMPAS in a criminal trial by a judge
- S2: Data curation process for getting the training and evaluation datasets for the algorithm
- S3: Model development process for COMPAS

Hazards:

- For S1:
  - H1: The judge does not understand or misunderstands the prediction. (L2, L3)
  - H2: The judge misuses the prediction by COMPAS. (L2, L3)
  - H3: The judge misuses the system.
- For S2:
  - H1: The curated data is not anonymized (appropriately) (L1, L4)
  - H2: The curated data is low-quality in terms of validity/correctness (L1, L2)
  - H3: The curated data does not cover specific racial groups. (L2, L3).
- For S3:
  - H1: The model cannot predict recidivism accurately (L1, L2, L3).
  - H2: The developers do not understand how the model makes its predictions (L1, L2).

Step 2: Create a control structure diagram

Mapping system dynamics

The second step involves drawing a diagram that shows the sub-systems within our system boundary and their connections. This diagram is called a control structure, and Leveson and Johnson define the key terms of the control structure as:

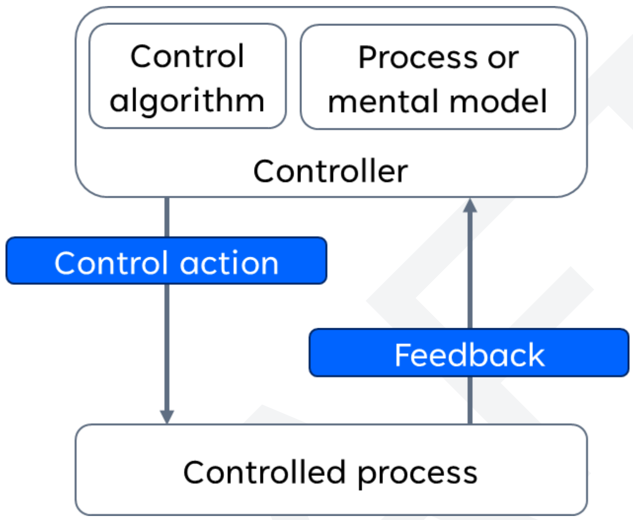


Figure 4. A simple control structure diagram with one controller and one controlled process

- **Control structure**: A hierarchical control structure is a system model that is composed of feedback control loops. An effective control structure will enforce constraints on the behavior of the overall system.
- A hierarchical control structure is composed of control loops. In general, a controller may provide **control actions** to control some processes and to enforce constraints on the behavior of the controlled process. A controller may be represented by a human decision-maker, automatic decision-making, or a combination thereof. Various constellations of human and automatic control exist, often expressed in terms of 'level of automation'. The **control algorithm** represents the controller's decision-making process—it determines the control actions to provide based on some inputs (information or a sensor signal).
- Controllers also have **process models** that represent the controller's internal beliefs used to make decisions. Process models may include beliefs about the process being controlled or other relevant aspects of the system or the environment. In a formal sense, they specify four conditions:
  - Goal condition: the objective and constraints of the controller
  - Action condition: the actions the controller has at its disposal to steer the controlled process to the objective while respecting the constraints

    - Observability condition: the real-time information the controller needs about the controlled process in order to determine the right course of action
    - Model condition: the broader understanding of the controlled process, as well as how the control structure acts on it
- **Mental models** are equivalent to process models for controllers that are executed by individuals or groups of people. If the control structure entails both human and automatic controllers, the human controller's mental model also includes the beliefs and understanding about the automatic controller and how it affects decision-making and the controlled process. Process models may be updated in part by **feedback** used to observe the controlled process.
- The control structure is hierarchical, meaning that the controllers at the top have a higher authority and control - and as we go down, the level of control and authority decreases.
- All the arrows down are control actions, arrows up are feedback, and arrows in and out of a box (horizontally) are the inputs and outputs. Input and output could signal information and communication that is passed between collaborators. However they are not control actions because they do not pose a constraint on behavior of the controlled process.

At its core, a systems lens for ML applications invites the developer to zoom out from the ML model and situate it in its context of use. The following are guidelines for developing a control structure for ML system. To develop a control structure diagram for a defined system boundary for an ML system:

- Examine and understand how the ML-based system works at a technical and operational level while understanding the context of use and the broad social system. Concretely, gaining this understanding will involve engaging with system developers, users, and a broader set of stakeholders.
- Identify stakeholders that are relevant to the given system boundary.
- Identify all the automated and technical components necessary for the functioning of the given system.
- Outline the responsibilities, key functions, and roles that individuals, groups, and technical component/sub-systems hold for all the elements you identified in the previous two steps. This step also relies on a sufficient understanding of the people, products, and processes involved in the ML development pipeline. Conduct the necessary research to acquire this information.
- Decide if the identified sub-systems (i.e., people, automated system, etc.) are controllers or controlled processes per the definition provided above.
- Determine the hierarchy between various controllers and controlled processes by deciding which sub-system is posing constraints on another sub-system.
- Identify the control actions and feedback (if present) after having the initial hierarchical arrangement.
- Outline the control algorithm and process or mental model for a given controller briefly.

Here is a non-exhaustive list of controllers, control actions, and feedback common to ML systems:

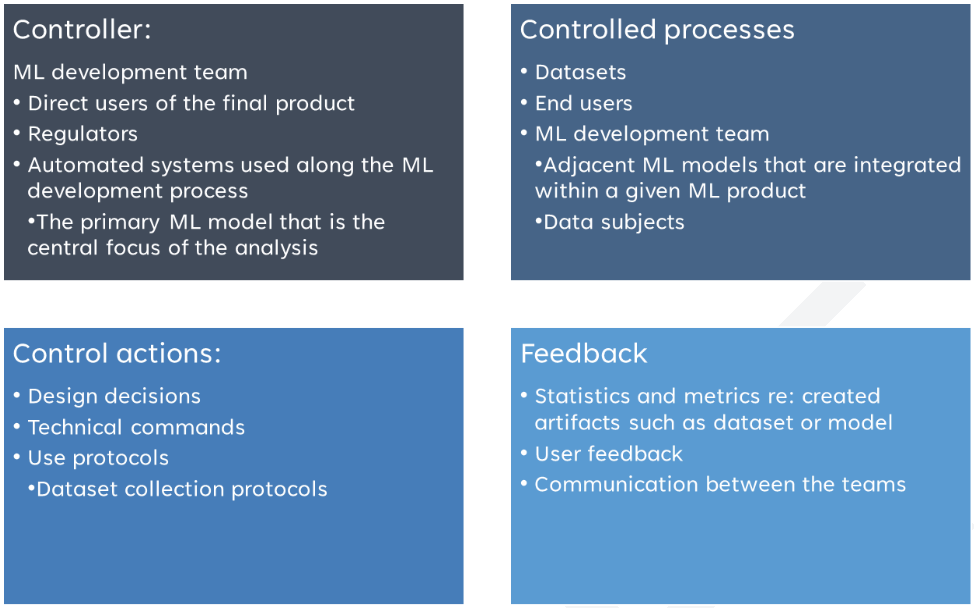
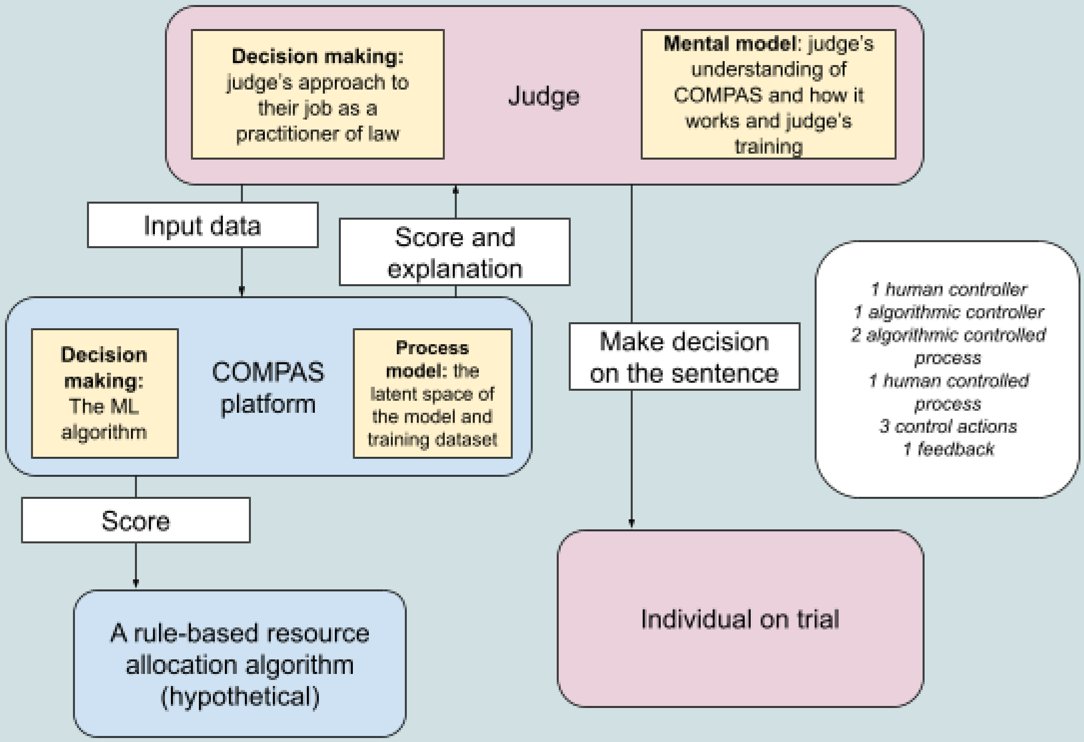


Figure 5. Potential examples of controller, controlled process, control action and feedback for ML systems

The most valuable part of drawing a control structure diagram for a system along the ML development pipeline is that it allows you to get a full perspective on how the different components of an ML pipeline relate to each other. It is well-known that it is impossible/very difficult to develop an abstraction of the complex relationships that exist between decision-makers, teams, automated systems, and end-users in the ML development/deployment process. In the first round of analysis, draw a simple diagram with 3 to 4 controllers and controlled processes that correspond to 4-5 control actions. Control structure diagrams are refined as the system and understanding of that system is developed.

Example:

Simplified control diagram for the use system boundary (S1)



Step 3: Identifying unsafe control actions

Examining unsafe interactions

From the control structure diagram, we use the control actions for step 3. Leveson defines the following key terms for this step:

- **Definition**: An Unsafe Control Action (UCA) is a control action that, in a particular context and worst-case environment, will lead to a hazard. For example, incorrect processing of training data could lead to potential discrimination against certain user groups. In this scenario, incorrect processing covers a broad range of unsafe control actions.
- **Control action can be unsafe in four different ways:**
  - Not providing the control action
  - Providing the control action or providing the incorrect control action
  - Providing a potentially safe control action but too early, too late, or in the wrong order
  - Providing the control action for too long or stopping it too soon (for continuous control actions, not discrete ones).

To develop the list of unsafe control action, compile the list of all the control actions identified in Step 2. For each control action:

- Review the list of hazards and losses from Step 1.
- Review your control structure from Step 2.
- Identify under what conditions could the control action be unsafe given the four different heuristics identified earlier. Think through the following questions:
  - Under what conditions, would **providing** this [CONTROL ACTION] lead to [HAZARD]?
  - Under what conditions, would **not providing or incorrectly providing** [CONTROL ACTION] lead to [HAZARD]?
  - Under what conditions, would **providing** this [CONTROL ACTION] **too early or too late** [IN A GIVEN TIMELINE] lead to [HAZARD]?
  - Under what conditions, would **providing** this [CONTROL ACTION] **too long or stopping it too soon** lead to [HAZARD]?

When thinking about the conditions for potentially unsafe control actions, examine the states of the controllers, controlled process, and controlled actions identified in Step 2. You have completed Step 3 when you have examined all the control actions from Step 2. The output of this step is a list of unsafe control actions that are linked to hazards identified in Step 1 and the control structure developed in Step 2.

A UCA contains five parts: source, type, control action, context, and link to hazard. The first part is the controller that can provide the control action. The second part is the type of unsafe control action (provided, not provided, too early or too late, stopped too soon or applied too long). The third part is the control action or command itself (from the control structure). The fourth part is the context discussed above, and the last part is the link to hazards (or sub-hazards). The same approach can be applied to human controllers to identify UCAs. The following diagram illustrates what a typical UCA table looks like for a PHASE analysis:

Controller	Controlled process	Control action	Context	Providing	Not providing	Too late or too early	Applied too long or stopped too early	Hazard
Classifier for filtering data	Dataset	Eliminate certain individuals as they are outliers	Recidivism algorithm in criminal justice	The filter eliminated too many data points.	The filter captured too many of the outliers, making it difficult to develop an accurate model.	The filter is applied before the data is cleaned up for missing values.	The filter keeps eliminating the data from continuous use of the system.	A low-quality training dataset is generated.

Notably, the ML development process - especially the model development and data processing is highly iterative. The developer of the system is generally rapidly iterating on the model architecture, choices of hyperparameters, etc. Furthermore, ML systems themselves are probabilistic in nature, and the control action that might be given from a given system could change over time. Furthermore, some generative systems develop emerging capabilities that could present themselves as emerging control actions. Therefore, to think through potential unsafe control actions, it is important to consider the unique properties of ML systems/development processes by carefully thinking through the *type* and *context* of the unsafe control action.

Example:

Controller	Controlled process	Control action	Context	Providing	Not providing	Too late or too early	Applied too long or stopped too early	Hazard
Judge	COMPAS platform	Input data	The judge inputs the required data into the COMPAS platform.	The judge enters incorrect information	The judge does not provide the key information	Not applicable	Not applicable	H3. The judge miuses the system.

Step 4: Identifying loss scenarios

Identifying potential causal scenarios

Once the UCAs are identified, in the last step, you will brainstorm and identify all the potential causal scenarios for each UCA. These are called loss scenarios - the circumstances that could potentially lead to a UCA, which could ultimately lead to a loss. To identify the loss scenarios:

- Examine all the components of a control structure diagram, including the exchange between different sub-systems and the inner workings of each system.
- Examine controller behavior by studying potential issues related to the controller by identifying 1) inadequate/incorrect controller algorithm or decision-making process, 2) inadequate/incorrect process or mental model.
- Examine potential avenues of inadequate feedback, which can mean that either feedback is not received or sent. Alternatively, the received feedback is not adequate or correct.

Each UCA generally corresponds to multiple loss scenarios. Typically it is relatively easy to identify at least 2 to 5 loss scenarios per UCA.

Example:

Unsafe control action	**Loss Scenarios**
The judge enters incorrect information	The judge's mental model of the COMPAS platform is incorrect and they do not know what data is needed for the COMPAS platform to work.
	The system does not provide any feedback about the quality of the data inputted by the judge. It just provides the predicted score.
The judge does not provide the key information.	The COMPAS platform does not have the means of inputting of all the data.
	The platform does not provide any feedback about the quality of the data inputted by the judge. It just provides the predicted score.

Now what: how do we synthesize the findings from PHASE?

After completing the first iteration of these guidelines on a given ML application, you will have the following:

- A list of system-level losses and hazards that are linked to each other
- A control structure diagram showing the key controllers, controlled processes, control actions and feedback loops
- A list of unsafe control actions
- A list of loss scenarios for each unsafe control action.

The next step is to synthesize and prioritize this information in the form of safety requirements, which outline the necessary constraints for ensuring that a given system is safe (i.e., will not lead to the losses identified for the given system). Development of safety requirements starts by closely examining the loss scenarios and seeing what types of constraints could minimize or eliminate the possibility of unsafe control actions in those scenarios. Sometimes, safety requirements are also oriented toward minimizing harm from an unsafe control action if it does occur. For ML systems, these requirements can be technical in nature (i.e., using or not using specific features in a dataset or using ML models optimized for fairness metrics), institutional (i.e. establishing specific product management and communication protocols between model and product development teams), or regulatory (i.e., requiring ML-based products to go through conformity assessments). Developing requirements is beyond the scope of this guideline, and it will not be a part of the assignment accompanying this guideline.

Case study: Predictive Algorithm in Clinical Decision Making

Overview of the case study

The use of predictive algorithms in clinical diagnosis is a common application of machine learning algorithms. In this case study, we perform STPA on an ML algorithm used to predict the likelihood of late-onset sepsis (LOS) in pre-term infants before a clinical suspicion. This algorithm could be used as an early warning system for clinicians to start early diagnostic testing for LOS and, ultimately, a quicker treatment. Clinicians could use this algorithm as a tool for “earlier than suspected detection of LOS in a group of very preterm infants.” This algorithm is used in a human-in-the-loop setting where a clinician will always need to make a decision based on the output of the algorithm. In this case study, the predictive algorithm is being developed at a hospital based in the Netherlands and is currently running in a shadow mode for research purposes where the clinicians are not actively using the system as part of their diagnosis.

Neonatal sepsis is a highly infectious illness that poses a major threat to the health and survival of newborns on a global scale. In particular, premature infants and infants with very low birth weight (VLBW) are susceptible to neonatal sepsis. Neonatal sepsis is typically classified based on when it occurs. Early onset sepsis (EOS) develops in the first few days of life and is primarily caused by “bacteria transmitted from the mother to the baby during labor or the prenatal period”. Late-onset sepsis (LOS) occurs after the first week of life and can be attributed to bacteria acquired from the baby's environment. The ramifications of late-onset sepsis (LOS) encompass neurological impairment, bronchopulmonary dysplasia, extended hospitalization, multi-organ system failure, and mortality. On a global scale, neonatal sepsis ranks as the third leading cause of neonatal deaths, accounting for 13% of them. Despite proactive treatments in high-income countries, 39% of all neonatal sepsis cases result in significant disability or death.

The timely diagnosis of LOS is difficult because its symptoms are quite comparable to other types of neonatal conditions. Furthermore, laboratory tests (ex. Positive blood culture) provide limited value as they illustrate the disease when it has progressed quite significantly. Considering the importance of early diagnosis and detection, clinicians often prescribe a range of antibiotics as they suspect an LOS diagnosis based on clinical signs. However, early prescription of antibiotics has its own negative side effects on infants. Most recently, clinicians have been looking at diagnosing LOS at an earlier time via other methods, including predictive ML algorithms.

Tags: #supervised learning, #human-in-the-loop, #research, #Europe

Approach to analysis

For this application of STPA, the lead analyst gathered the necessary information from published articles and conversations with two experts who were involved in building this ML system. Two advisors also provided feedback on the drafts of the STPA analysis and this feedback was incorporated for the final analysis. The three main scopes that were examined for this analysis include **1) data collection and processing, 2) model development and evaluation and 3) clinical implementation**. The focus of the data collection and processing was to develop the appropriate training dataset for model development that has a positive clinical impact. The model development and evaluation stage considered key factors such as model interpretability, including features that detect actionable insights about the patient's well-being and creating model outputs that have potential use in clinical practice. Even though the clinical implementation of this project is not fully developed, in this analysis, we use the simulated clinical use as the baseline for our analysis.

Summary of findings

Scope 1 - Data collection and processing

Step 1: Identify purpose of analysis

Losses:
The main loss for this system is **loss of life or health (L1)** to the preterm infants, which could result of late diagnosis of LOS or early prescription of antibiotics in case of misdiagnosis. Within the medical context, other common losses include **loss of time (in terms of efficiency) (L2)** and **loss of reputation/credibility (L3)** for the physicians, clinicians and the hospital in charge of a case. Considering the regulations around medical data, another key element to consider is the **loss of patient privacy (L4)**. Lastly **loss of quality (L5)** is of value for clinicians (in terms of care) and development team (for technology).

Stakeholders: patient (infants), parents (caretakers/guardians), physicians and fellow clinicians (i.e. nurses), hospital management, research and development team
Values: patient health and safety, privacy, reputation, quality in delivery of care, quality in delivery of technology, efficiency,

ID	**Loss**
L1	Loss of life/injury
L2	Loss of time
L3	Loss of repulation/credibility
L4	Loss of patient privacy
L5	Loss of quality

System boundary: The focus of the data collection and processing was to develop the appropriate training dataset for model development that has a positive clinical impact.

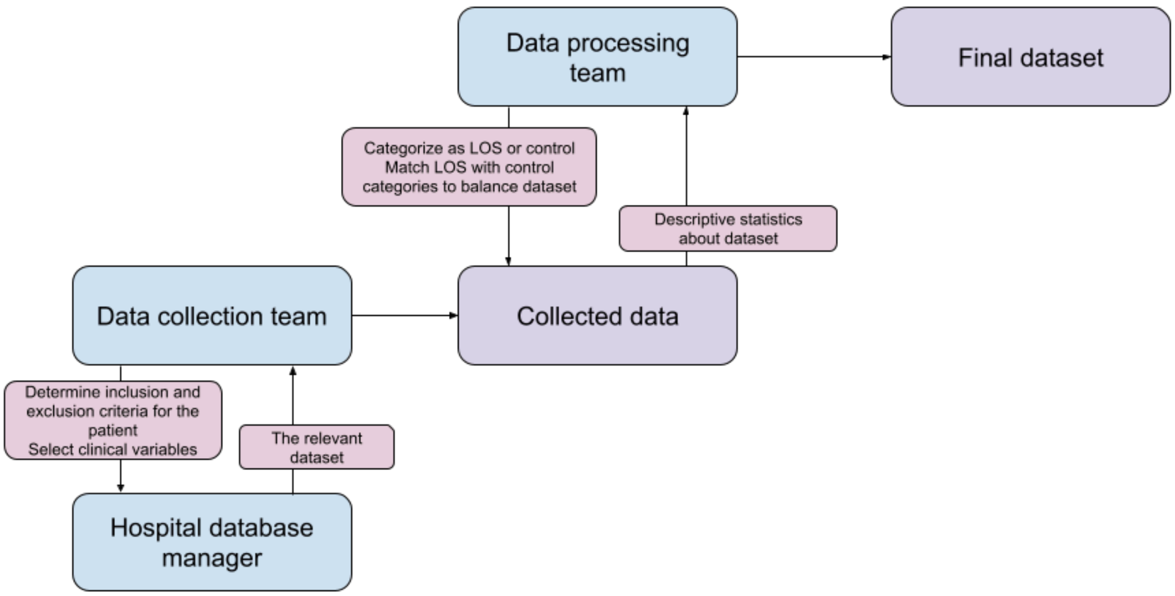
System- level hazards:

- H1. The system creates a training dataset that does obide by the relevant data privacy regulations.
- H2. The system creates a training dataset that is unbalanced across relevant labels.
- H3. The system creates a training datasets that is mislabeled.
- H4. The system creates a training dataset training set contains features from clinical data that are exist due to physicians suspicion of LOS.
- H5. The system creates at training dataset that is too small for training a model training

		Associated Losses				
ID	**Hazards**	L1	L2	L3	L4	L5
H1	The system creates a training dataset that does obide by the relevant data privacy regulations	FALSE	FALSE	TRUE	TRUE	TRUE
H2	The system creates a training dataset that is unbalanced across relevant labels.	TRUE	TRUE	TRUE	FALSE	TRUE
H3	The system creates a training datasets that is mislabeled.	TRUE	TRUE	TRUE	FALSE	TRUE
H4	The system creates a training dataset training set contains features from clinical data that are exist due to physicians suspicion of LOS.	TRUE	FALSE	TRUE	FALSE	TRUE
H5	The system creates at training dataset that is too small for model training .	FALSE	TRUE	FALSE	FALSE	TRUE

Step 2: Create control structure diagram

The control structure diagram consists of three human controllers and 2 artifacts that are controlled and output/inputs of other controllers. Based on the information gathering, the data collection and processing scope did not include any automated processes.



The three human controllers include 1) the data collection team, 2) the hospital database manager, and 3) the data processing team. This scope does not include other relevant actors, such as clinicians or the individuals who are in charge of ensuring that data privacy protocols are followed within a hospital research team. We focus on analyzing four control actions for the next step, which include 2 control actions between the data collection team and hospital database manager and 2 others between the data processing team and collected data. It is noteworthy that the data processing and collection team consisted of the same people for this project.

Step 3: Identify unsafe control actions

We brainstormed a preliminary list of nine unsafe control actions following the original STPA heuristics. The UCAs lead to at least one hazard. The unsafe control actions were focused on the different mistakes and issues that could happen when selecting the relevant data from the hospital database and processing it to prepare the training/evaluation dataset. These include but are not limited to 1)not following the inclusion criteria or having inappropriate inclusion criteria), 2) selecting improper clinical variables to be included as features in the collected dataset, 3) miscategorizing the collected dataset as LOS or control, and 4) balancing the dataset improperly.

Step 4: Identify loss scenarios

We uncovered four overarching loss scenarios, including 1) Data collection/processing team did not engage with the necessary stakeholders (i.e. physicians) to understand what type of data needs to be collected or how the data needs to be processed, 2) Improper inspection/feedback/descriptive statistics about the data being collected or processed, 3) Inadequate data anonymization practice in the hospital, and 4) Below the standard approaches to data processing (i.e poor data balancing)

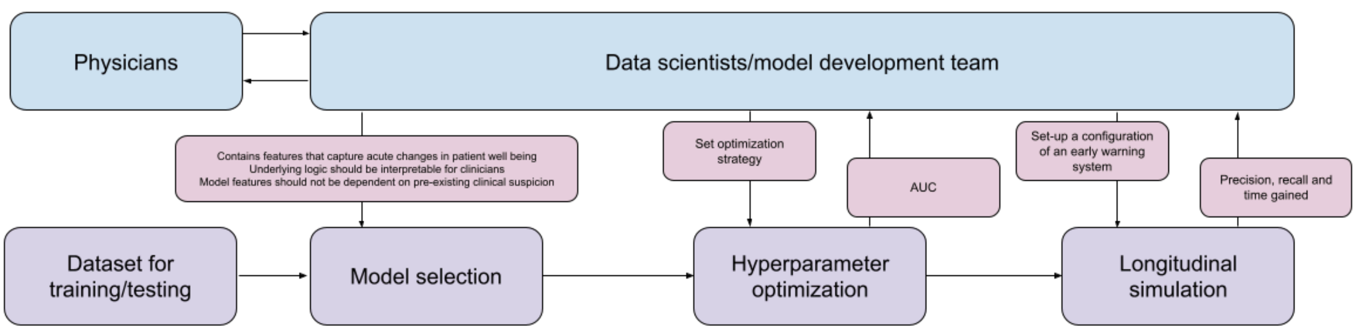
Scope 2 - Model development and evaluation

Step 1: Identify purpose of analysis
Losses: The main loss for this system is **loss of life or health (L1)** to the preterm infants, which could result of late diagnosis of LOS or early prescription of antibiotics in case of misdiagnosis. Within the medical context, other common losses include **loss of time (in terms of efficiency) (L2)** and **loss of reputation/credibility (L3)** for the physicians, clinicians and the hospital in charge of a case. Considering the regulations around medical data, another key element to consider is the **loss of patient privacy (L4)**. Lastly **loss of quality (L5)** is of value for clinicians (in terms of care) and development team (for technology). **Stakeholders:** patient (infants), parents (caretakers/guardians), physicians and fellow clinicians (i.e. nurses), hospital management, research and development team **Values:** patient health and safety, privacy, reputation, quality in delivery of care, quality in delivery of technology, efficiency, **ID** **Loss** L1 Loss of life/injury L2 Loss of time L3 Loss of repulation/credibility L4 Loss of patient privacy L5 Loss of quality **System boundary:** The model development and evaluation stage considered key factors such as model interpretability, including features that detect actionable insights about the patient's well-being and creating model outputs that have potential use in clinical practice **System- level hazards:** ● H1. The system creates a model/output that does not meet performance requirements (ex. accuracy, precision and recall). ● H2. The system creates a model/output that is not interpretable. ● H3. The system does not evaluate adequately/correctly for performance of the model and output. ● H4. The system does not evaluate adequately/correctly for interpretablity the model and output. ● H5. The system does not evaluate adequately/correctly for privacy and fairness the model and output.

		Associated Losses				
ID	**Hazards**	L1	L2	L3	L4	L5
H1	The system creates a model/output that does not meet performance requirements (ex. accuracy, precision and recall).	TRUE	TRUE	TRUE	FALSE	TRUE
H2	The system creates a model/output that is not interpretable.	TRUE	TRUE	TRUE	FALSE	TRUE
H3	The system does not evaluate adequately/correctly for performance of the model and output.	TRUE	FALSE	TRUE	FALSE	TRUE
H4	The system does not evaluate adequately/correctly for interpretablity the model and output.	TRUE	FALSE	TRUE	FALSE	TRUE
H5	The system does not evaluate adequately/correctly for privacy and fairness the model and output.	TRUE	FALSE	TRUE	TRUE	TRUE

Step 2: Create control structure diagram

The control structure consists of 2 human sub-systems and 4 technological artifacts/processes. The main controller here is the data scientist or the full model development team. This controller collaborates with the physician as indicated by the input/output arrows and has control over three main process/artifacts of model selection hyperparameter optimizationa and longitudinal simulation. Dataset for training and testing is an input to the model selection process. The output of all the purple boxes is an input for the next one.



Step 3: Identify unsafe control actions

The STPA identified twelve Unsafe Control Actions (UCAs) within the model development and evaluation process, focused on three key areas: model selection, hyperparameter optimization, and longitudinal simulation. Unsafe actions included failing to provide clinically meaningful features, relying on biased or implausible data, applying misaligned optimization strategies, and simulating model behavior under unrealistic assumptions. Many UCAs stemmed from poor timing (e.g., decisions made without clinician input), incorrect content (e.g., label leakage), or insufficient oversight in how configurations were defined and tested. These control flaws often reflect breakdowns in feedback loops, incomplete mental models of the clinical environment, or misaligned priorities between model performance and real-world usability.

Step 4: Identify loss scenarios

The loss scenarios observed in the model development and evaluation scope reveal how system-level failures can emerge from control structure breakdowns. Many losses stemmed from missing feedback loops—for example, when model developers made architectural or optimization decisions without timely clinician input, leading to models that were uninterpretable or clinically misaligned. In other cases, loss resulted from incorrect mental models held by developers, such as overestimating model generalizability or misunderstanding how features function in practice. This led to the inclusion of misleading or biased inputs and evaluation strategies that failed to reflect real clinical needs.

Flawed process models also played a role: simulations configured with idealized data or assumed clinician behavior failed to account for the complexity and variability of real-world care. This misrepresentation affected how model outputs were interpreted and trusted, degrading quality (L5) and credibility (L3). Additionally, poorly designed control algorithms, such as hyperparameter optimization routines that prioritized aggregate accuracy over clinically meaningful trade-offs, resulted in models that underperformed in early detection or treated different patient subgroups unfairly—exacerbating risks to life (L1), time (L2), and equity (L5).

Overall, these scenarios demonstrate that technical choices made during model development are deeply shaped by system control dynamics. Without accurate process models, bidirectional feedback, and control logic that reflects clinical goals, even high-performing models can lead to serious downstream harm.

Scope 3 - Clinical implementation

Step 1: Identify purpose of analysis

Losses:
The main loss for this system is **loss of life or health (L1)** to the preterm infants, which could result of late diagnosis of LOS or early prescription of antibiotics in case of misdiagnosis. Within the medical context, other common losses include **loss of time (in terms of efficiency) (L2)** and **loss of reputation/credibility (L3)** for the physicians, clinicians and the hospital in charge of a case. Considering the regulations around medical data, another key element to consider is the **loss of patient privacy (L4)**. Lastly **loss of quality (L5)** is of value for clinicians (in terms of care) and development team (for technology).

Stakeholders: patient (infants), parents (caretakers/guardians), physicians and fellow clinicians (i.e. nurses), hospital management, research and development team
Values: patient health and safety, privacy, reputation, quality in delivery of care, quality in delivery of technology, efficiency,

ID	**Loss**
L1	Loss of life/injury
L2	Loss of time
L3	Loss of repulation/credibility
L4	Loss of patient privacy
L5	Loss of quality

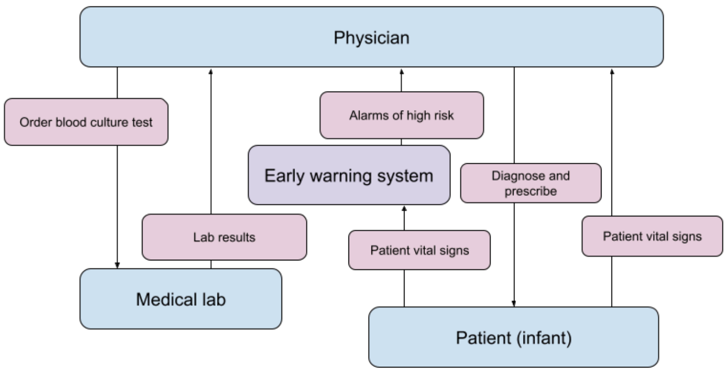
System boundary: Use the predictive algorithm to deliver quality patient care and treat or order necessary tests as early as possible.

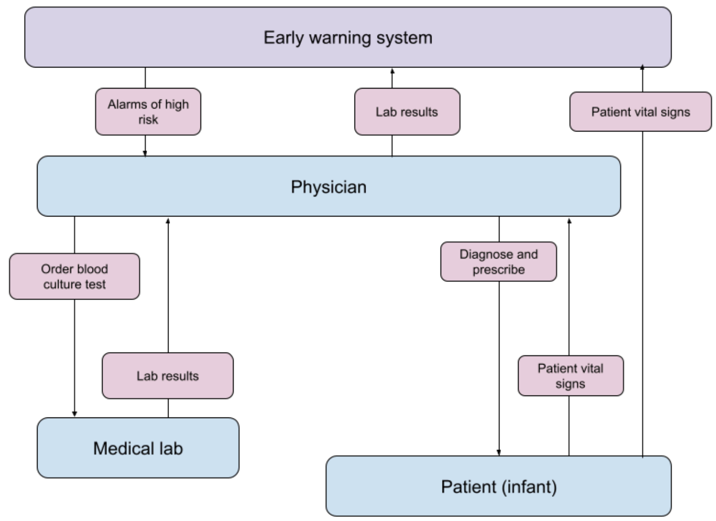
System- level hazards:
- H1. The physician does not understand or misunderstands the output of the predictive algorithm.
- H2. The physician misdiagnosis for LOS and prescribes antibiotics.
- H3. The physican diagnosis LOS too late.

		Associated Losses				
ID	**Hazards**	L1	L2	L3	L4	L5
H1	The physician does not understand or misunderstands the output of the predictive algorithm.	TRUE	TRUE	TRUE	FALSE	TRUE
H2	The physician misdiagnosis for LOS and prescribes antibiotics.	TRUE	FALSE	TRUE	FALSE	TRUE
H3	The physican diagnosis LOS too late.	TRUE	FALSE	TRUE	FALSE	TRUE

Step 2: Create control structure diagram

The control diagram has three human/group sub-systems and one automated system. Depending on whether the physicians have to use the alarm system or not the set-up of the control hierarchy could be different. In the left control diagram the physician is the highest level of controller because they do not need to use the early warning system. They just get feedback from it in form of alarms. Whereas, in the right scenario the physicians are required to use the early warning system and there the system is higher in the hierarchical control structure. The control actions are different in each case. The case to the left has 2 main control actions as opposed to three control actions for the control diagram to the right. The nurse and the parent/caretaker will likely be involved in this scope but for now we have kept them outside for simplicity of the analysis.





Step 3: Identify unsafe control actions

In our STPA of a predictive early warning system for Late-Onset Sepsis (LOS) in preterm infants, we identified twelve Unsafe Control Actions (UCAs) across three key control relationships:

1. **Physician → Medical Lab (Order Blood Culture Test):**
   UCAs occur when the physician fails to order a test when needed, orders it unnecessarily, delays it, or orders it prematurely. These unsafe actions can stem from misinterpretation of risk signals or overreliance on the algorithm, leading to missed diagnoses or over-testing.

2. **Physician → Patient (Diagnose and Prescribe):**
   Unsafe actions include failing to diagnose or prescribe treatment when LOS is present, doing so prematurely or based on misinterpretation, or delaying action. These behaviors are particularly risky given the narrow treatment window for LOS.

3. **Early Warning System → Physician (Alarm of High Risk):**
   UCAs arise when the system fails to generate an alarm when risk is high, raises a false alarm, delays the alarm, or issues too many alerts. These failures can result in missed interventions or alarm fatigue, undermining clinician trust

and response behavior.

Step 4: Identify loss scenarios

The loss scenarios identified in this STPA reflect systemic vulnerabilities that can be traced to control flaws in feedback, process models, and control algorithms — all of which shape the behavior of both human and automated components in the system.

- **Missed or delayed detection of LOS** (e.g., the physician fails to act or the system fails to alarm) often stems from missing or delayed feedback loops between the early warning system and the physician. If the alarm is not triggered or is delayed, the physician's process model may not reflect the actual physiological state of the infant, leading to underreaction and loss of life or health (L1).

- **Unnecessary testing or treatment** (e.g., false positives or premature test orders) frequently result from an **incorrect control algorithm** in the early warning system or a mismatched mental model on the part of the physician. If the algorithm over-predicts risk or the physician does not correctly interpret the model's output, it can lead to inefficient use of resources (L2), harm from overtreatment (L1), and reduced trust in both the physician and the system (L3, L5).

- **Delayed diagnosis or prescription** may be caused by a misalignment between the physician's process model and the true system state. If the physician assumes low risk due to a lack of alerts or unclear decision thresholds, their timing of intervention may not match the needs of the patient, resulting in harm and reputational loss.

- **Alarm fatigue and clinician desensitization** arise when the control algorithm lacks contextual awareness or adaptive thresholds, leading to frequent false or low-value alarms. This breakdown in the control loop between the system and the user causes clinicians to ignore or disable alerts, effectively breaking the feedback channel and increasing the likelihood of missed true positives (L1, L3, L5).

- **Breakdowns in clinician-system coordination** often reflect incomplete or ambiguous system feedback, where alerts or outputs lack actionable guidance, or are not well-integrated into clinical workflow. These issues suggest a need for better designed process models within both the system and the training clinicians receive on how to use it.

Case study: RL use in Insulin Injection

Overview of the case study

In this application of STPA, we are investigating potential hazards from using an automated insulin injection system that uses a reinforcement learning algorithm. Insulin injection is a critical part of treating patients with diabetes. Currently, the appropriate dosage of insulin and other drugs is studied in clinical trials. It is well-reported that considering the inherent limitations/requirements in clinical trials, only a limited number of people can participate in clinical trials. Often, people with comorbidities and from minority populations do not meet participation requirements. The appropriate dosage determined from the clinical trials is generally optimal for people who are most similar to the group that got to participate in the trials. This poses a health risk to people who are not part of this group. Considering the limitations posed by clinical trials and other adjacent issues the medical community is exploring the use of RL algorithms for personalization of drug dosage as one type of a solution.

Currently, insulin injection is currently being done manually, where the patient is responsible for monitoring their glucose level after meals and injecting on a needed basis. Alternatively, patients with the appropriate socioeconomic means use an artificial pancreas, which facilitates insulin injection automatically. Currently, the algorithmic system used in artificial pancreas does not use machine learning algorithms. The motivation behind using RL algorithms in insulin injection is to personalize, optimize, and fine-tune the amount of insulin dose that is needed for each individual person based on their specific reaction (i.e. change in glucose level) to what they eat and their environmental conditions. To illustrate what we could learn from an application of STPA for such a system, we consider three different scopes of analysis, including **1) Implementation of RL algorithm in an artificial pancreas, 2) Design of RL algorithm, 3) Use of RL-based artificial pancreas in patient care**.

Tags: #reinforcement learning, #human-out-of-the-loop, #research, #Canada

Approach to analysis

The analysis was led by the lead author, who collaborated with two experts in reinforcement learning. One of the analysts focused on developing RL algorithms for insulin injection and had worked extensively with physicians in the field to understand the clinical setting and impact of this technology on potential users. The three analysts met on a bi-weekly basis from February 2023 to Aug 2023 and worked on two different STPA analyses following the steps outlined in the STPA handbook (Leveson 2016), which was supplemented by the lead author's expertise in applying the STPA process. It is important to note that use of RL in insulin injection is at the research stage, and the analysis did not consider operational settings for implementing such a technology. The focus of this analysis was to examine how the STPA process applies to ML systems that are based on RL algorithms.

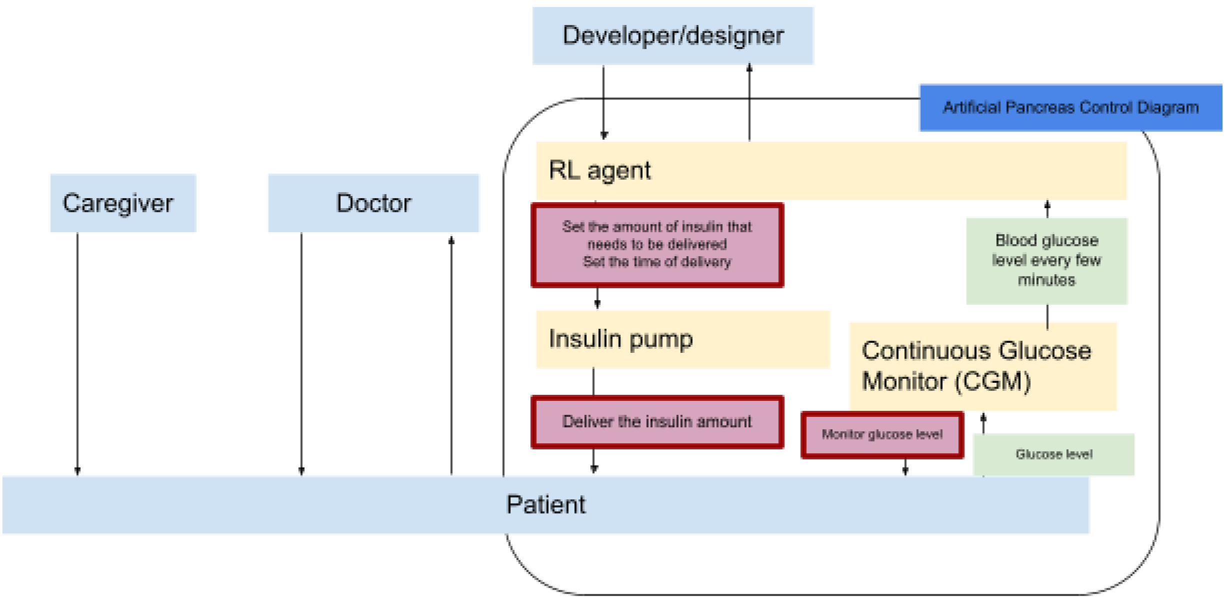
Summary of findings

Scope 1 - Implementation of RL algorithm in an artificial pancreas

Step 1: Identify purpose of analysis
Losses: From the stakeholders and their values, some of the key losses for this case study are 1) Loss of life/injury, 2) Loss of quality of life, 3) Loss of efficiency, 4) Loss of money. **Stakeholders:** We identified the patient, principal caregiver, primary physician, specialist physician, artificial pancreas developer, the researcher responsible for developing the RL algorithm, sensor manufacturer, regulating body for medical devices (Health Canada or FDA), provincial government (who might be subsidizing such a device) and healthcare administration (of a hospital or a clinic). Each of these stakeholder hold a different set of values **Values:** Patients value a healthy life where they have a sense of autonomy, a good quality of life, and the ability to care for their condition (diabetes) efficiently. They also want to minimize the cost of taking care of their condition. The primary caregiver also cares the health and quality of life of the patient and also desires time, cost and effort efficiency. Furthermore, they want to preserve their own mental health and well-being as they are often worried/concerned about the patient. The primary physician needs to and wants to maintain professional medical practice while optimizing the time and effort required to manage a patient's condition. They also care for the patient's health and quality of life. In addition, they want to maximize their income. **ID** **Loss** L1 Loss of life/injury L2 Loss of quality of life (efficiency, autonomy, amount of effort) L3 Loss of efficiency L4 Loss of money **System boundary:** For the given scope, the goal of the system is to deliver the right dosage of insulin at the needed time for all patients. **System- level hazards:** 1) Artificial pancreas provides a larger dose than needed at a given time to the patient. 2) Artificial pancreas provides a smaller dose than needed at a given time to the patient .

ID	Hazards	Associated Losses L1	L2	L3	L4
H1	Artificial pancreas provides a larger dose than needed at a given time to the patient.	TRUE	TRUE	TRUE	TRUE
H2	Artificial pancreas provides a smaller dose than needed at a given time to the patient .	TRUE	TRUE	TRUE	TRUE

Step 2: Create control structure diagram



When modelling the control structure diagram we identified 3 automated controllers (RL agent, insulin pump and continous glucose monitor) and one human controller (patient). The patient is not actively taking part of controlling any part of this process – they are the receiver of decisions made by the RL agent and provide of data through the CGM. There are four main control actions that we examined in this scope including 1) RL agent to pump: set the amount of insulin that needs to be delivered,2) RL agent to pump: set the time of delivery and 3) Pump to patient: deliver the insulin amount 4) CGM to patient Monitor glucose level. There is no direct feedback loop from the patient to the pump or the RL agent. The CGM scans the patient's glucose level and sends that to the RL agent.

Step 3: Identify unsafe control actions

The STPA analysis identified 17 Unsafe Control Actions (UCAs), primarily focused on functional performance hazards: either too much or too little insulin being delivered. The key UCAs reflect breakdowns in coordination among automated components (RL agent, insulin pump, CGM) and lack of human-in-the-loop oversight. The four critical control actions considered were:

1. RL Agent → Pump: Set insulin amount
2. RL Agent → Pump: Set delivery time
3. Pump → Patient: Deliver insulin
4. CGM → RL Agent: Monitor glucose level

Examples of UCAs include: ● RL agent sets too large a dose or delays insulin delivery, causing hypo- or hyperglycemia (linked to Hazard 1 or 2). ● The pump delivers insulin at the wrong time or fails to deliver, due to incorrect or missing commands. ● CGM provides inaccurate glucose data, which misguides the RL agent's decision-making. Importantly, no UCAs in this scope directly related to social or ethical risks were identified. Instead, the risks were tied to automated functional failure, stemming from missing or incorrect control actions, inadequate timing, or inaccurate sensor data.
Step 4: Identify loss scenarios
After examining potential controller behaviours and existing feedback loops, we brainstormed an initial list of 7 loss scenarios that apply across numerous UCAs. The losses include 1) Improper design and training of RL algorithm, 2) Incorrect observations due to a faulty CGM, 3) Improper design of the reward function, 4) Pump, valves or tubes in the system are physically damaged and not functioning properly, 5) The pump controller is out of tune with the input from the RL agent., 6) CGM is physically broken, 7) The CGM controller loop is sampling at a higher or a lower frequency than expected. These loss scenarios draw attention to potential design and manufacturing issues of the various controllers.

Scope 2 - Design of RL model

Step 1: Identify purpose of analysis

Losses:
From the aforementioned stakeholders and their values, some of the key losses for this case study are 1) Loss of life/injury, 2) Loss of quality of life, 3) Loss of efficiency, 4) Loss of money.

Stakeholders: We identified the patient, principal caregiver, primary physician, specialist physician, artificial pancreas developer, the researcher responsible for developing the RL algorithm, sensor manufacturer, regulating body for medical devices (Health Canada or FDA), provincial government (who might be subsidizing such a device) and healthcare administration (of a hospital or a clinic). Each of these stakeholder hold a different set of values

Values: Patients value a healthy life where they have a sense of autonomy, a good quality of life, and the ability to care for their condition (diabetes) efficiently. They also want to minimize the cost of taking care of their condition. The primary caregiver also cares the health and quality of life of the patient and also desires time, cost and effort efficiency. Furthermore, they want to preserve their own mental health and well-being as they are often worried/concerned about the patient. The primary physician needs to and wants to maintain professional medical practice while optimizing the time and effort required to manage a patient's condition. They also care for the patient's health and quality of life. In addition, they want to maximize their income.

ID	**Loss**
L1	Loss of life/injury
L2	Loss of quality of life (efficiency, autonomy, amount of effort)
L3	Loss of efficiency
L4	Loss of money

System boundary: Design and train an RL model which performs well with simulated clinical data and environment.

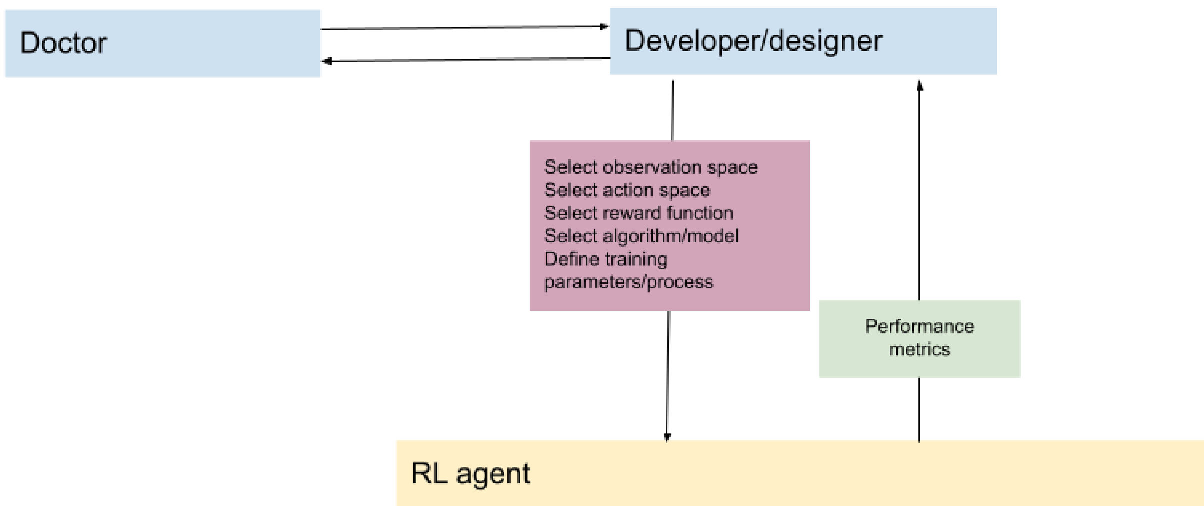
System- level hazards:

1. RL model provides a larger dose than needed at a given time (in simulation environment) .
2. RL model provides a smaller dose than needed at a given time (in simulation environment) .

		Associated Losses			
ID	**Hazards**	L1	L2	L3	L4
H1	RL model provides a larger dose than needed at a	TRUE	TRUE	TRUE	TRUE
H2	RL model provides a larger dose than needed at a	TRUE	TRUE	TRUE	TRUE

Step 2: Create control structure diagram

We have kept this diagram relatively simple. The main control is in the developer/designer of the system where they make the design decisions for the RL agent. There are no other ML models that are being used to make this decision for the system. The doctor provides feedback to the developer based on their expertise in their collaboration.



Step 3: Identify unsafe control actions

In Scope 2, Unsafe Control Actions center around the design decisions made by the developer of the RL model used in an artificial pancreas. These UCAs occur during the specification of the reward function, construction of the training environment, selection of input features (state representation), tuning of hyperparameters, and decisions about when and how to evaluate the model. Each control action, when performed incorrectly or omitted, can lead to either over- or underestimation of insulin needs in simulated environments, which poses downstream risks once deployed. All UCAs were tied to two functional hazards: the RL model recommending a larger or smaller dose than required. Notably, UCAs here reflect design-stage control flaws that are less visible in real-time operation but crucially shape system safety.

Step 4: Identify loss scenarios

The identified loss scenarios in Scope 2 emphasize how design flaws in the RL model can propagate into clinical harm. For example, an improperly specified reward function may prioritize short-term glucose drops at the expense of long-term patient health, while a narrow or biased training environment may exclude data from underrepresented populations, leading to ineffective dosing policies. Other scenarios include premature model evaluation, unstable convergence due to poorly tuned exploration parameters, or exclusion of key physiological indicators in the input state. Many of these scenarios stem from missing clinical feedback, biased assumptions about patient diversity, and insufficient validation procedures. Although these issues arise in simulation, they carry forward to deployment—making design-time decisions a critical site for hazard mitigation.

Scope 3 - Use of RL-based artificial pancreas in patient care.

Step 1: Identify purpose of analysis

Losses:
From the aforementioned stakeholders and their values, some of the key losses for this case study are 1) Loss of life/injury, 2) Loss of quality of life, 3) Loss of efficiency, 4) Loss of money.

Stakeholders: We identified the patient, principal caregiver, primary physician, specialist physician, artificial pancreas developer, the researcher responsible for developing the RL algorithm, sensor manufacturer, regulating body for medical devices (Health Canada or FDA), provincial government (who might be subsidizing such a device) and healthcare administration (of a hospital or a clinic). Each of these stakeholder hold a different set of values

Values: Patients value a healthy life where they have a sense of autonomy, a good quality of life, and the ability to care for their condition (diabetes) efficiently. They also want to minimize the cost of taking care of their condition. The primary caregiver also cares the health and quality of life of the patient and also desires time, cost and effort efficiency. Furthermore, they want to preserve their own mental health and well-being as they are often worried/concerned about the patient. The primary physician needs to and wants to maintain professional medical practice while optimizing the time and effort required to manage a patient's condition. They also care for the patient's health and quality of life. In addition, they want to maximize their income.

ID	**Loss**
L1	Loss of life/injury
L2	Loss of quality of life (efficiency, autonomy, amount of effort)
L3	Loss of efficiency
L4	Loss of money

System boundary: The physician need to prescribe and manage/monitor a proper diabetes management plan for the patient. The family also needs to help with the management.

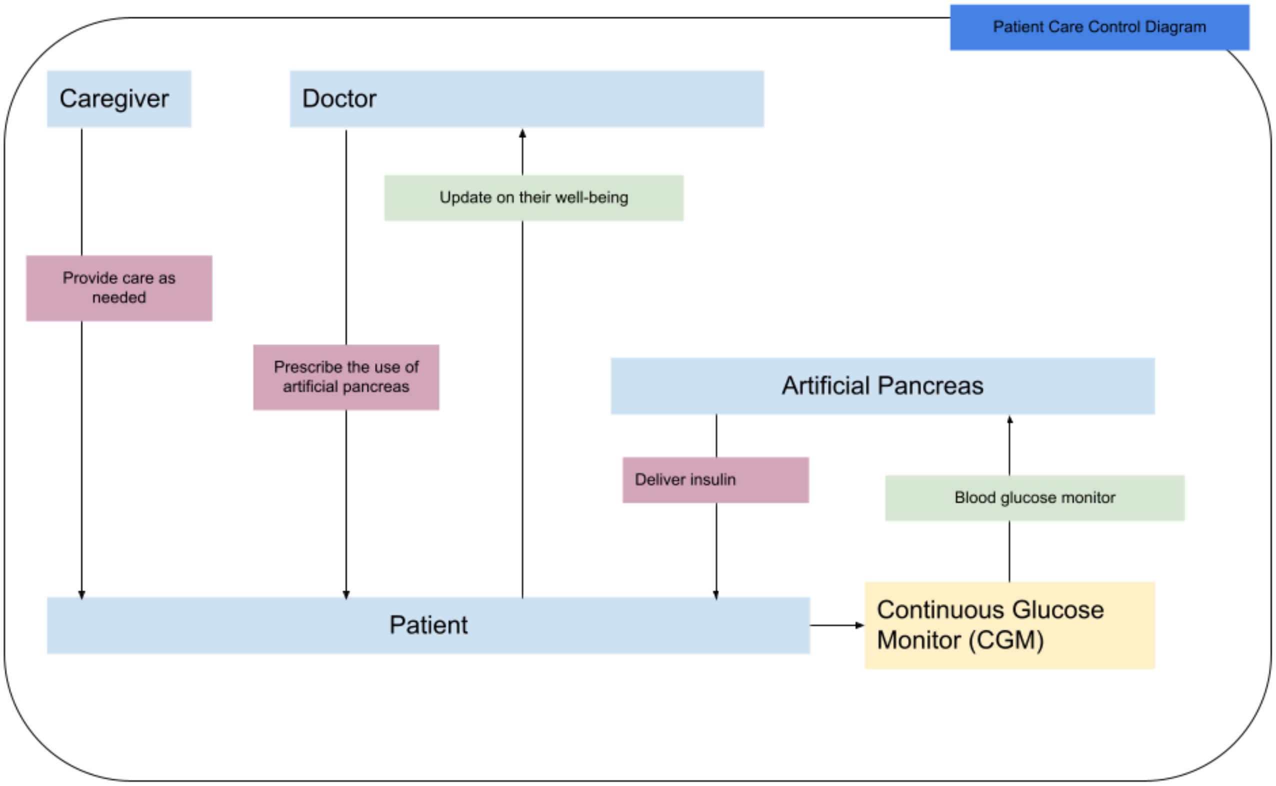
System- level hazards:

3. Physician misdiagnoses the patient's condition.
4. Physician cannot monitor the patient's status.

		Associated Losses			
ID	**Hazards**	L1	L2	L3	L4
H1	Physician misdiagnoses the patient's condition.	TRUE	TRUE	TRUE	TRUE
H2	Physician cannot monitor the patient's status.	TRUE	TRUE	TRUE	TRUE

Step 2: Create control structure diagram

There are four main components - three sub-systems are humans and one is the artificial pancreas. All are higher in the control hierarchy compared to the patient. The care giver is likely lower in control hierarchy than what is actually illustrated here.



Step 3: Identify unsafe control actions

In Scope 3, the UCAs focus on the interaction between human and automated agents in real-world deployment. Three key control actions were analyzed: (1) caregivers providing care to patients, (2) physicians prescribing the artificial pancreas (AP), and (3) the AP delivering insulin. Unsafe control actions included caregivers failing to intervene or acting incorrectly, physicians prescribing the device to unsuitable patients or avoiding its use when beneficial, and the AP delivering insulin inappropriately or not at all. These UCAs highlight failures not only in functional performance but also in trust, training, and the social infrastructure surrounding device use. Both system-level hazards—misdiagnosis and lack of monitoring—were triggered by these unsafe control actions, particularly when feedback between components was missing or ineffective.

Step 4: Identify loss scenarios

The loss scenarios in Scope 3 reflect how poor coordination, device failures, and human misunderstandings can lead to serious health outcomes. For example, caregivers may misinterpret device alerts or overlook malfunctions, leading to incorrect or missing interventions. Physicians may over-rely on or avoid recommending the AP, causing mismatches between patient needs and technology capabilities. On the technical side, the RL-based artificial pancreas may misdeliver insulin due to incorrect input interpretation or physical failures like sensor detachment or battery depletion. These scenarios reveal how fragile system safety becomes when multiple human and technical actors interact without robust safeguards or shared understanding. While the RL model might perform well in simulation, these real-world loss scenarios emphasize the need for user training, system monitoring, and contextual awareness to ensure safe and equitable care.

Case study: T2I use in creative practice

Overview of the case study

Applying STPA at three stages of the ML development pipelines enabled analysts to uncover a wide range of failures and hazards without the need for detailed information on the specific T2I model. In particular, the three scopes of analysis (training data processing, product integration, and end use) each allowed practitioners to look beyond a specific model in isolation and focus on processes and interactions as it is integrated with other actors and technical systems. In the next three sections, we reflect on the failures and hazards discovered at each stage. The analysis uncovered known failures and hazards (e.g., the creation of non-consensual sexual imagery) and novel ones that have not been recognized to the best of our knowledge (e.g., the impact of English-word filters on lexical change). We also identified failures and hazards that could present social and ethical risks for different actors, which can inform the prioritization of mitigation strategies.

Approach to analysis

STPA could be applied to various scopes of analysis that require a different set of information. Scoping an ML system for such an analysis is a non-trivial task. An ML system can be divided into its component parts (e.g., training data, model, user interface), development process (e.g., training, testing, early deployment), stakeholders involved (e.g. ML developers, a community of developers interfacing and building on the model APIs, end-users), and so on. The results of the analysis can be drastically different depending on the chosen scope of analysis [10, 42].

Based on the information gathered from the interviews with T2I experts and workshops with artists, we identified three scopes of analysis along the ML development pipeline:

- Scope 1: the data processing necessary for creating a training dataset
- Scope 2: how a T2I model is integrated into a production environment along with other ML models
- Scope 3: how an artist uses T2I model demo as part of their creative process

Selecting the scopes along the ML development pipeline allows for the examination of critical processes and interactions as discussed by previous scholarship [31 , 67 , 70]. We chose to focus on these scopes of analysis because we had access to the most amount of publicly available and shareable information about the elements involved.

The STPA analysis was led by the first author of this paper and three of the co-authors provided feedback on iterations of the analysis. A separate analysis was conducted for each one of the scopes. In total three STPA analyses were conducted. The lead author spent somewhere between 10 - 12 hours implementing the STPA process on one scope. All sources of data were used as input for STPA process. The STPA analysis resulted in a list of unsafe control actions and corresponding safety requirements.

We followed the original STPA process for each one of the scopes (as described below) and did not alter them to specifically uncover social and ethical risks.

Summary of findings

Scope 1 - data processing

Step 1: Identify purpose of analysis
Losses: From the key values brought up by artists, software engineers, and responsible ML practitioners, we formulated the following losses: **(L1) Loss of creativity; (L2) Loss of diversity; (L3) Loss of accessibility; (L4) Loss of efficiency; (L5) Loss of quality; and (L6) Loss of reputation.** **Stakeholders:** From our interviews, we identified three key stakeholders involved in data processing: (1) software engineers who prepare the dataset for training; (2) responsible ML practitioners who provide guidelines on what should or should not be included in the dataset; and (3) lawyers who consult on privacy and legal requirements for datasets. The software engineers collaborate closely with the practitioners responsible for developing the model. Lawyers provide legal requirements (i.e., intellectual property and privacy) for training datasets to the responsible ML and the product team. There is limited information available on how data processing is done for T2I models. For our proof-of-concept analysis, we use the data processing steps outlined in the publicly available data card for the Parti model [17]. The steps outlined in this data card are similar to what other T2I model creators have discussed in their publications [63, 68] **Values:** Artists emphasized the values of fostering creativity, serving a diverse audience, accessibility of artistic mediums to a diverse group of artists, efficiency in their creative process, and preservation of the artist's reputation/identity. The developers and evaluators of T2I models emphasized the value of creating efficient systems that generate appropriate and quality images in response to a given text. They also emphasized the value of diversity (i.e. the importance of serving a diverse audience of users with their models) and the importance of protecting their team and their company's reputation. **ID** **Loss** L1 Loss of creativity L2 Loss of diversity in generated image L3 Loss of accessibility (who can use these systems) L4 Loss of efficiency (in generating image) L5 Loss of quality L6 Loss of reputation **System boundary:** In our application, we bound the data processing stage to start from the identification of one or multiple source datasets and end with the creation of a training dataset ready for model development purposes. In this scope, we do not directly analyze the steps involved in how the data sources were obtained and who was involved in the data collection process. The focus is on how the data sources are processed for creating a training dataset. **System-level hazards:**

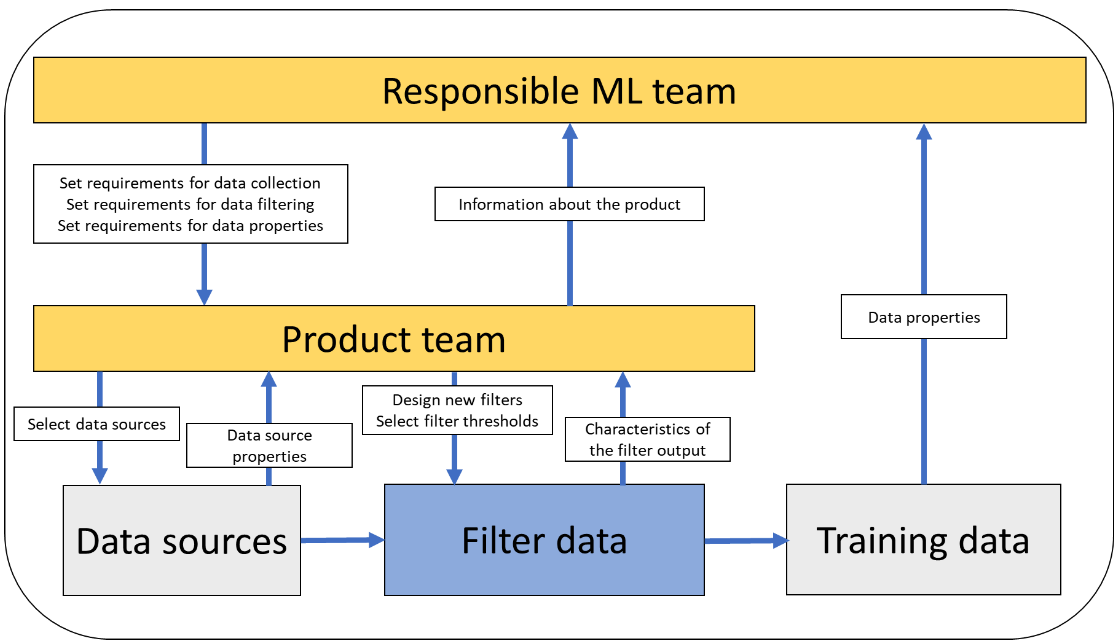
The next STPA step is to identify hazards by considering these losses in relation to the previously identified goal of training data processing. We developed three potential hazards:

- **H1:** System is unable to create a training dataset that contains good quality text-image pairs.
- **H2:** System is unable to create a training dataset that does not contain harmful content.
- **H3:** System is unable to create a training dataset with a diverse representation.

		Associated Losses					
ID	**Hazards**	L1	L2	L3	L4	L5	L6
H1	System is unable to create a training dataset that contains good quality text-image pairs	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE
H2	System is unable to create a training dataset that does not contain harmful content	FALSE	TRUE	TRUE	FALSE	FALSE	TRUE
H3	System is unable to create a training dataset with a diverse representation	TRUE	TRUE	TRUE	FALSE	FALSE	TRUE

Step 2: Create control structure diagram

Then, we model the control structure. A potential control structure configuration for the data processing stage, as illustrated in Figure 3, includes two organizational/human controllers and one automated controller. We selected the "responsible ML team" and "product team" as human controllers because they can make key decisions in the data processing scope. For this analysis, we assume the responsible ML team is in charge of determining the parameters for a good training dataset and identifying key ethical and legal considerations. The product team is responsible for developing the models that underpin the T2I demo artists will use. We selected as "filter data" an automated controller to reflect where all of the automated filtering operations take place. We identified 6 control actions, denoted in the boxes with down arrows in Figure 3.



For our analysis, we focused on 4 of the 6 control actions occurring between the three controllers and data sources including 1) Set requirements for data filtering, 2) Set requirements for dataset properties, 3) Design new filters and 4) Select filter thresholds. We chose to focus on these 4 control actions because they specifically focus on data processing as opposed to source data collection.

Step 3: Identify unsafe control actions

The STPA analysis for the training data processing stage identified 22 potential Unsafe Control Actions (UCAs) that could lead to three key system hazards: (H1) low-quality text-image pairs, (H2) inclusion of harmful content, and (H3) lack of diverse representation in the training dataset. These UCAs span both human-to-human coordination and human-to-automation interactions, revealing a range of sociotechnical vulnerabilities in the data processing pipeline.

A central pattern in the UCAs involved coordination breakdowns between the responsible ML team and the product team. Multiple UCAs describe cases where the responsible ML team either does not provide, or provides too late, the requirements for data filtering and dataset properties. These lapses were linked to *all three hazards*, highlighting the importance of internal timelines and communication infrastructure for safety and fairness outcomes.

On the automation side, the product team's design and parameterization of data filters—particularly for non-English content, adult content, and other sensitive categories—were also common sources of unsafe actions. Overly aggressive filters risked excluding a disproportionate amount of data, leading to dataset sparsity and reduced representational diversity. Conversely, under-performing filters failed to catch inappropriate or harmful material, raising risks to both system performance and user safety.

Importantly, several UCAs also revealed underlying structural assumptions in technical design, such as an over-reliance on English-speaking content or narrow definitions of harmful material. These UCAs demonstrate how implicit values embedded in filtering logic can shape the resulting training data—and by extension, the outputs of the T2I system—in ways that undermine creativity, accessibility, and equity.

Collectively, the 22 UCAs show how ethical and functional risks emerge not just from model outputs, but from upstream decisions and omissions during data preparation. The STPA framework enabled the identification of these risks at the organizational and procedural levels, offering a practical method for embedding safety and responsibility into ML data pipelines from the outset.

Step 4: Identify loss scenarios

The STPA analysis surfaced a set of plausible loss scenarios that reveal how unsafe control actions—especially those involving coordination breakdowns between human actors—can lead to hazards across quality, safety, and equity dimensions in T2I training data.

A recurring theme across these scenarios is misalignment in communication and timelines between the responsible ML team and the product development team. For instance, when the product team fails to clearly articulate the intended use case and end-user context, the responsible ML team must make assumptions or use generic guidelines. This can result in filters or dataset requirements that are misaligned with the actual context of use—potentially leading to loss of diversity (L2) or loss of accessibility (L3) if, for example, culturally specific or marginalized representations are omitted due to inappropriate filtering criteria.

Another class of scenarios involves delayed or incomplete feedback loops. In several cases, the responsible ML team is overloaded with concurrent review requests and cannot deliver requirements on time. As a result, the product team may move forward with data processing using default or outdated filters, leading to datasets that include harmful content (H2) or lack diverse representation (H3). In reverse, if the product team withholds contextual information until late in the process, the responsible ML team cannot meaningfully tailor safeguards.

These loss scenarios also highlight missing or inadequate policies for identifying appropriate requirements. In the absence of established guidelines, the responsible ML team may resort to ad hoc solutions or default to internal brainstorming—introducing inconsistency and reducing rigor in content filtering. This contributes to losses of quality (L5) and reputation (L6) if the resulting dataset enables problematic model outputs.

Finally, many scenarios emerge from an overreliance on assumed defaults—e.g., assuming that product timelines always take precedence, or that ML teams can generalize across use cases. These assumptions break down in high-stakes or sensitive applications, revealing the need for explicit timing coordination, shared mental models of responsibilities, and sustained investment in process-level safety measures.

Together, these loss scenarios underscore how sociotechnical failures—such as poor internal communication or unclear governance—can propagate through seemingly mundane development processes, with significant downstream consequences for T2I model safety and trustworthiness

Scope 2 - model integration

Many AI ethics assessments and audits focus on one ML model [25 , 76]. However, in production, multiple models often are used to achieve the intended functions for a given product or demo [83].We conducted STPA to examine the integration of a T2I model in a productionized demo, such as those released by Stability AI [2] and OpenAI [63]. From our interviews with T2I developers and existing literature [54 , 64], we identified demos often include at least three types of models: (1) an input prompt classifier (which either block or filter the text prompt), (2) the T2I model, and (3) output image classifiers (which either block or filter the generated image). For any given T2I demo, there could be multiple classifiers employed for filtering text prompts and images. For the purposes of this illustrative analysis, we assume there are only two classifiers. One for "adult" content run on the input text prompt and one for analyzing the generated image.

Step 1: Identify purpose of analysis
Losses: From the key values brought up by artists, software engineers, and responsible ML practitioners, we formulated the following losses: **(L1) Loss of creativity; (L2) Loss of diversity; (L3) Loss of accessibility; (L4) Loss of efficiency; (L5) Loss of quality; and (L6) Loss of reputation.** **Stakeholders:** From our interviews, we identified three key stakeholders involved in data processing: (1) software engineers who prepare the dataset for training; (2) responsible ML practitioners who provide guidelines on what should or should not be included in the dataset; and (3) lawyers who consult on privacy and legal requirements for datasets. The software engineers collaborate closely with the practitioners responsible for developing the model. Lawyers provide legal requirements (i.e., intellectual property and privacy) for training datasets to the responsible ML and the product team. There is limited information available on how data processing is done for T2I models. For our proof-of-concept analysis, we use the data processing steps outlined in the publicly available data card for the Parti model [17]. The steps outlined in this data card are similar to what other T2I model creators have discussed in their publications [63, 68] **Values:** Artists emphasized the values of fostering creativity, serving a diverse audience, accessibility of artistic mediums to a diverse group of artists, efficiency in their creative process, and preservation of the artist’s reputation/identity. The developers and evaluators of T2I models emphasized the value of creating efficient systems that generate appropriate and quality images in response to a given text. They also emphasized the value of diversity (i.e. the importance of serving a diverse audience of users with their models) and the importance of protecting their team and their company’s reputation. **ID** **Loss** L1 Loss of creativity L2 Loss of diversity in generated image L3 Loss of accessibility (who can use these systems) L4 Loss of efficiency (in generating image) L5 Loss of quality L6 Loss of reputation **System boundary:** Here the goal of the system is to create a T2I demo for public and creative use.

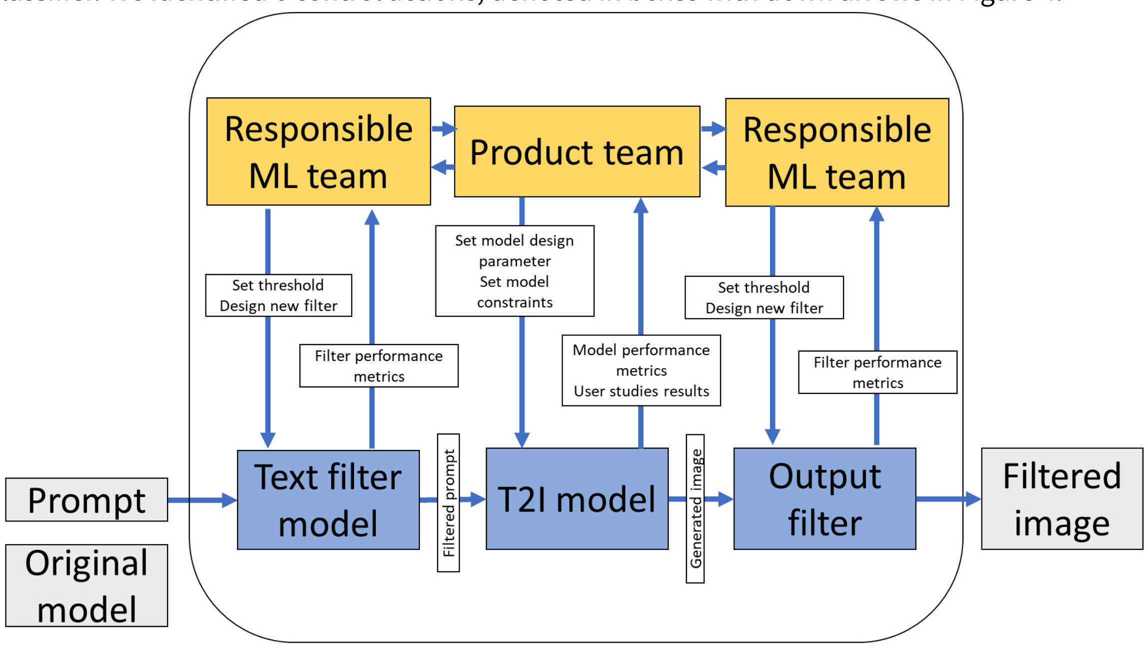
System-level hazards:

- H1: System creates an image that does not match the prompt.
- H2: System cannot generate an image.
- H3: System generates an unsafe image (i.e., with adult content).

		Associated Losses					
ID	Hazards	L1	L2	L3	L4	L5	L6
H1	System creates an image that does not match the prompt.	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE
H2	System cannot generate an image.	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE
H3	System generates an unsafe image	FALSE	TRUE	TRUE	FALSE	TRUE	TRUE

Step 2: Create control structure diagram

A potential configuration of the control structure (as illustrated in Figure 4) for this scope of analysis includes two organizational/ human controllers and three automated controllers. The organizational/human controllers are the "responsible ML team" and the "product team." The automated controllers are the input text classifier, the T2I model, and the output image classifier. We identified 6 control actions, denoted in boxes with down arrows in Figure 4.



Step 3: Identify unsafe control actions

The STPA for Scope 2 identified 16 Unsafe Control Actions (UCAs) related to the integration of multiple machine learning models within a T2I demo. These UCAs spanned both organizational processes and technical configurations across a system composed of at least three models: an input prompt classifier, the T2I model itself, and an output image classifier.

Many UCAs emerged from coordination breakdowns between the responsible ML team and the product team, such as delays in setting model constraints, unclear or incomplete filtering requirements, and misaligned timelines. For example, a UCA where the product team fails to implement appropriate filter thresholds for text prompts can result in the model rejecting legitimate prompts, thereby contributing to H2: system cannot generate an image or H3: generation of unsafe content.

Other UCAs centered around incorrect or poorly calibrated model parameters. These include scenarios where inappropriate constraints are placed on the T2I model version used in the demo, or where the responsible ML team designs overly restrictive filters that limit creative expression. These actions often had dual consequences: they reduced the system's ability to fulfill its intended creative purpose (H1: mismatch between prompt and image) and introduced ethical or reputational risks (L1: loss of creativity, L6: loss of reputation).

Critically, several UCAs revealed missing feedback loops—for instance, the product team does not receive feedback from the output image classifier, leading to a misalignment between model behavior and system goals. These gaps highlight structural limitations in how model integration decisions are made and updated across teams, particularly when responsibilities for safety and performance are fragmented.

The 16 UCAs illustrate how integrating multiple models into a cohesive product introduces unique hazards not visible when evaluating any one model in isolation. These include emergent risks at the intersection of classifier behavior, model constraints, and organizational workflow—risks that conventional single-model assessments or static checklists may fail to capture.

Step 4: Identify loss scenarios

The loss scenarios identified in Scope 2 demonstrate how failures in coordination, timing, and contextual understanding across organizational teams and model components can result in unsafe, inaccessible, or low-quality T2I outputs. These scenarios reflect not only technical missteps but also broader sociotechnical breakdowns in how safety, creativity, and usability are managed in product development.

A recurring pattern in the scenarios involves reuse or misconfiguration of pre-existing filter thresholds. For example, when the responsible ML team (RMT) fails to update thresholds for input or output filters (Scenarios 1 and 7), old parameters—possibly tuned for different use cases—are applied inappropriately. This leads to mismatches between the system's behavior and the actual user context, contributing to hazards such as the generation of irrelevant or blocked images (H1, H2) and losses like L1: creativity and L6: reputation. These failures stem from limited time, unclear ownership of filter configurations, or lack of technical knowledge for modifying embedded components.

Another critical set of scenarios centers on inappropriate filtering logic—either due to poor validation (Scenario 2) or misalignment with user intent (Scenario 4). In these cases, filters either fail to capture problematic content or overly restrict valid, expressive inputs. For instance, filters might exclude prompts common in marginalized communities or artistic experimentation, leading to L2: loss of diversity and L3: loss of accessibility.

Several scenarios illustrate how timing failures between teams undermine safety and effectiveness. For instance, when RMT delivers new prompt or image filters too late (Scenarios 3 and 8), the product team must launch with incomplete or outdated safeguards. These timing mismatches are often due to cycle time constraints, unclear prioritization, or overloaded review queues—problems deeply embedded in organizational workflows.

Finally, loss scenarios associated with model parameter setting and constraint definition reveal how cautious over-restriction by the product team (e.g., limiting colors or shapes, as in Scenario 6) may constrain user creativity, leading to L1, L4, and L5 losses. These decisions are often made out of legal or reputational concerns but may inadvertently alienate user groups or limit the system's artistic utility.

Together, these loss scenarios show that integration-stage hazards emerge from more than just technical flaws—they stem from misaligned expectations, insufficient feedback loops, and embedded organizational trade-offs. These failures compromise not only functional performance but also the system's capacity to serve diverse users in creative and socially responsible ways.

Scope 3 - use

This scope of analysis focuses on how an artist would use a T2I demo as part of their creative practice. The information we gathered from the artist workshops heavily informs this scope. To perform this illustrative STPA analyses, we assume the artist is a filmmaker, and they are creating a video for a client. The key stakeholders include the client, the filmmaker, and the public/viewer of the video. The scope focuses on how the filmmaker uses the T2I demo as a tool for generating images in their storyboard mock-up, where they visualize and share ideas with the client

Step 1: Identify purpose of analysis

Losses:
From the key values brought up by artists, software engineers, and responsible ML practitioners, we formulated the following losses: **(L1) Loss of creativity; (L2) Loss of diversity; (L3) Loss of accessibility; (L4) Loss of efficiency; (L5) Loss of quality; and (L6) Loss of reputation.**

Stakeholders: From our interviews, we identified three key stakeholders involved in data processing: (1) software engineers who prepare the dataset for training; (2) responsible ML practitioners who provide guidelines on what should or should not be included in the dataset; and (3) lawyers who consult on privacy and legal requirements for datasets. The software engineers collaborate closely with the practitioners responsible for developing the model. Lawyers provide legal requirements (i.e., intellectual property and privacy) for training datasets to the responsible ML and the product team. There is limited information available on how data processing is done for T2I models. For our proof-of-concept analysis, we use the data processing steps outlined in the publicly available data card for the Parti model [17]. The steps outlined in this data card are similar to what other T2I model creators have discussed in their publications [63, 68]

Values: Artists emphasized the values of fostering creativity, serving a diverse audience, accessibility of artistic mediums to a diverse group of artists, efficiency in their creative process, and preservation of the artist's reputation/identity. The developers and evaluators of T2I models emphasized the value of creating efficient systems that generate appropriate and quality images in response to a given text. They also emphasized the value of diversity (i.e. the importance of serving a diverse audience of users with their models) and the importance of protecting their team and their company's reputation.

ID	**Loss**
L1	Loss of creativity
L2	Loss of diversity in generated image
L3	Loss of accessibility (who can use these systems)
L4	Loss of efficiency (in generating image)
L5	Loss of quality
L6	Loss of reputation

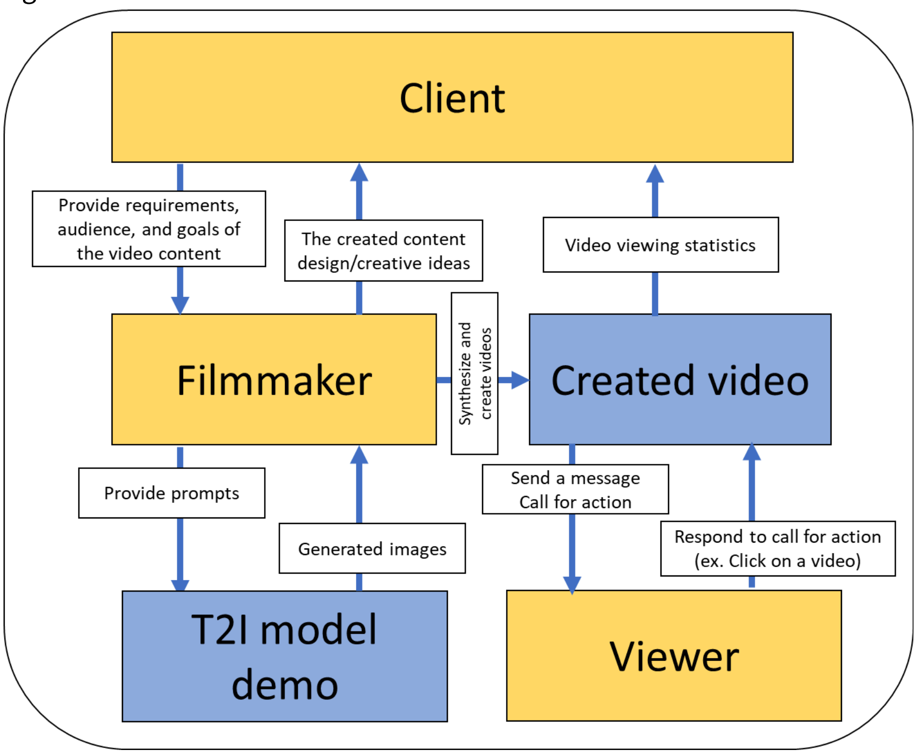
System boundary: The goal of our system is to support an artist in creating a video for a client using a T2I demo.

System-level hazards:
- H1: The artist cannot create a video. (L1, L3, L6)
- H2: The artist cannot create a video that meets client requirements. (L1, L3, L5, L6)
- H3: The video disseminates false and harmful information. (L2, L3, L5, L6)

ID	Hazards	Associated Losses					
		L1	L2	L3	L4	L5	L6
H1	The artist cannot create a video	TRUE	FALSE	TRUE	FALSE	FALSE	TRUE
H2	The artist cannot create a video that meets client requirements.	TRUE	FALSE	TRUE	FALSE	TRUE	TRUE
H3	The video disseminates false and harmful information.	FALSE	TRUE	TRUE	FALSE	TRUE	TRUE

Step 2: Create control structure diagram

A potential configuration of the control structure (illustrated in Figure 5 includes three human controllers (i.e., the client, filmmaker, and the viewer) and two automated controllers (i.e., the T2I model demo and the created video). For the purpose of this analysis, we focus on the 3 control actions between the 3 human controllers and the T2I model demo (denoted in boxes with down arrows in Figure 5): (1) Provide requirements/goals/audience for the video content; (2) provide prompts; (3) send a message.



Step 3: Identify unsafe control actions

The analysis identified 9 Unsafe Control Actions across interactions between the client, filmmaker, viewer, and T2I system. These UCAs reflect both communication failures and limitations in system behavior that constrain creative expression or introduce harm.

Several UCAs involved miscommunication or misalignment of expectations between the client and filmmaker. These include vague or missing project goals, which result in a video that misrepresents the client's intent or fails to meet their standards (linked to H2). Others reflect the challenges artists face in generating effective prompts for T2I systems—either due to a lack of domain alignment (e.g., the system can't interpret the prompt) or model bias, which disproportionately affects artists

from marginalized backgrounds (H1, H2, H3). Additional UCAs occurred during the viewer's interaction with the final product. Inaccurate or harmful interpretations of the video (H3) can emerge due to ambiguous visuals or misinformation generated by the T2I model. Viewers may also fail to respond or engage if the message is unclear or unrelatable (H1, H3). Overall, the UCAs point to critical sociotechnical gaps—from prompt literacy and accessibility to interpretability and equity—that can undermine the effectiveness and safety of T2I systems in real-world creative workflows.
Step 4: Identify loss scenarios
Each UCA was linked to 1–2 plausible loss scenarios that illustrate how control failures translate into real-world consequences. These scenarios highlight the cascading nature of misalignment, bias, and ambiguity in human-AI collaboration. Some scenarios show how unclear client communication leads the filmmaker to produce a misaligned storyboard, eroding trust and causing reputational or financial loss (L5, L6). Others emphasize how model limitations or prompt misalignment prevent the filmmaker from generating relevant images, resulting in lower-quality outputs and creative inefficiencies (L1, L4). Importantly, several scenarios spotlight equity issues—such as when artists from marginalized backgrounds receive worse output quality due to biased model training. These harms are not merely technical failures but represent deeper structural issues of accessibility, diversity, and inclusion in ML model development (L2, L3). Finally, viewer-side scenarios reveal that lack of transparency or clarity in the AI-generated visuals can lead to misinterpretation or misinformation spreading, causing ethical and reputational risks for both the artist and the client (L6).